

Case Study: Satellite Remote Sensing & Data Analytics for Crop Health Monitoring

<u>Project Title:</u>	Multi-Temporal Sentinel-2 Crop Health Assessment & Spatial Stress Detection
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<u>Tools & Tech:</u>	QGIS 3.x (GIS Mapping), R Programming (ggplot2, dplyr).
<u>Data Source:</u>	Sentinel-2 Satellite Multispectral Imagery (ESA Copernicus - Level 2A).
<u>Resolution:</u>	10 Meters / Pixel (Bands 4 & 8).

1. Project Background & Problem Statement:

- **The Real-World Problem:**

Farmers and agronomists traditionally walk through fields to check crop health. This manual survey method is slow, expensive, and fails on large land areas. More importantly, when crops show visible damage (like leaves turning yellow), significant crop loss has already occurred.

- **The Solution:**

By using **free satellite imagery** and **data analytics**, we can detect crop stress, water shortage, and nutrient deficiencies weeks before they become visible to the naked eye. This allows farmers to apply water or fertilizers only where needed, saving time and money.

2. Data Source & Technical Tools:

Data Source:

- **Platform:** European Space Agency (ESA) – Copernicus Open Access Hub / Sentinel Hub.
- **Satellite:** Sentinel-2 (Multispectral Instrument - MSI).
- **Bands Used:**

- **Band 4 (Red):** Measures chlorophyll absorption.
- **Band 8 (Near-Infrared / NIR):** Measures leaf cell density and healthy canopy structures.
- **Resolution:** 10 meters per pixel.

Software Stack:

- **QGIS:** To process satellite bands, calculate map algebra, and create spatial vegetation maps.
- **R Programming:** To run statistical tests (T-tests/ANOVA) and build clear visualizations (ggplot2, dplyr).

3. Step-by-Step Workflow & Methodology:

- **Satellite Data Acquisition:** Downloaded cloud-free Sentinel-2 imagery for two different dates (Peak Growing Stage vs. Maturation Stage).
- **Map Algebra in QGIS:** Used the Raster Calculator to compute the **Normalized Difference Vegetation Index (NDVI)**:

$$NDVI = \frac{Band\ 8\ (NIR) - Band\ 4\ (Red)}{Band\ 8\ (NIR) + Band\ 4\ (Red)}$$

- **Spatial Data Extraction:** Sampled pixel values across different field zones into a CSV file.
- **Data Analytics in R:** Cleaned the raw grid points in R, plotted distribution curves, and tested if the health shift between the two dates was statistically significant.

4. Results & Visual Outputs:

A. Geospatial Map Analysis (QGIS Output):

The calculated NDVI maps revealed distinct spatial health patterns across the field:

- **High Health (NDVI > 0.6):** Healthy, dense green canopy in central irrigated plots.
- **Low Health / Stress (NDVI < 0.3):** Clear stress pockets along field borders due to moisture drop or soil issues.



Figure 1: *QGIS Spatial Map showing vegetation vigor and chlorophyll distribution across field zones.*

NDVI Range	Vegetation Health Class	Agronomic & Spatial Interpretation
< 0.10	Non-Vegetated / Bare Soil	Open tracks, water channels, or unplanted soil patches.
0.10 – 0.30	High Stress / Sparse	Severe moisture stress, nutrient deficiency, or early crop emergence.
0.30 – 0.55	Moderate Canopy Vigor	Partial canopy closure, developing foliage.
0.55 – 0.85	Dense & Healthy Canopy	Active photosynthesis, full canopy closure in central irrigated plots.

B. Statistical Data Analytics (R Outputs):

To back up the visual raster maps with quantitative rigor, extracted pixel values were evaluated in R:

1. Biomass Density & Frequency Distribution:

The continuous density distribution plot highlights a complete multimodal shift in crop biomass between seasons. In March, pixel values peak around 0.38, whereas by July, the primary density mass shifts toward high chlorophyll values (> 0.75).

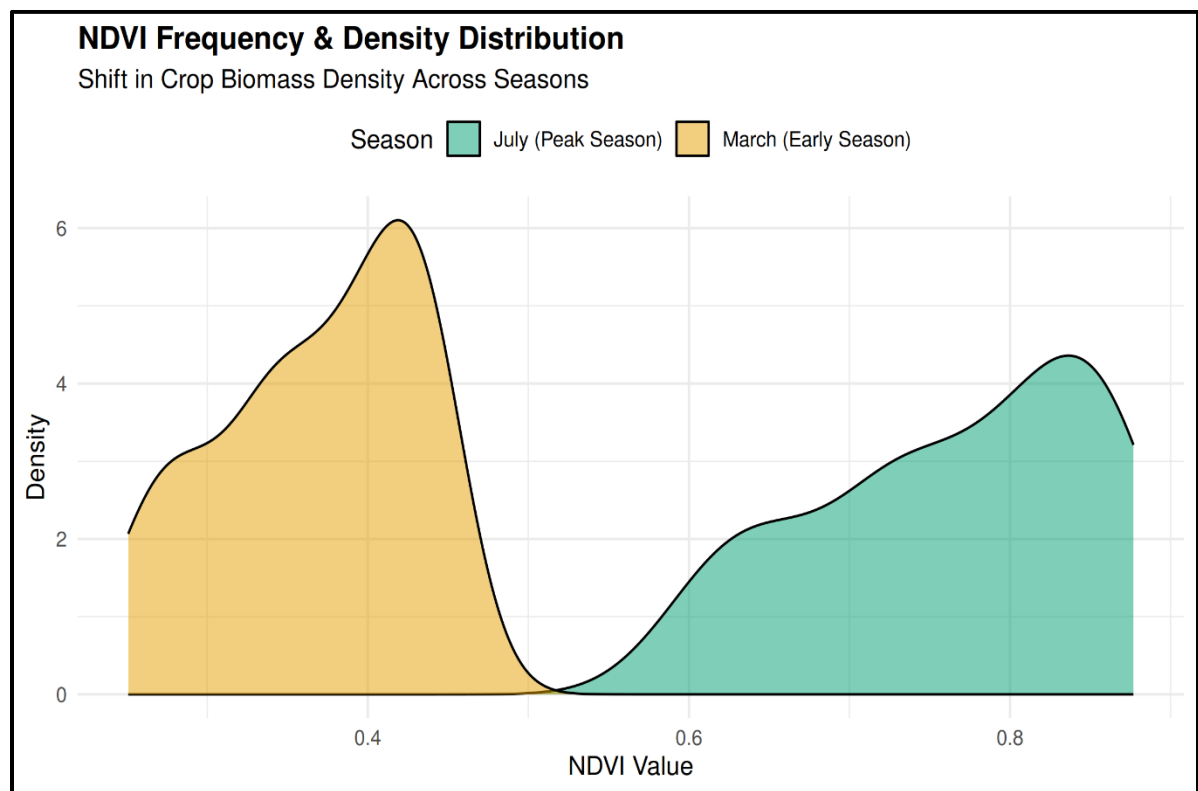


Figure 2: *Density Distribution Curve showing the shift in crop biomass density across seasons.*

2. Range & Dispersion of Seasonal Crop Vigor:

The boxplot illustrates the explicit variation and median jump in canopy density between the early growth phase (March Median ≈ 0.38) and peak vegetative performance (July Median ≈ 0.78).

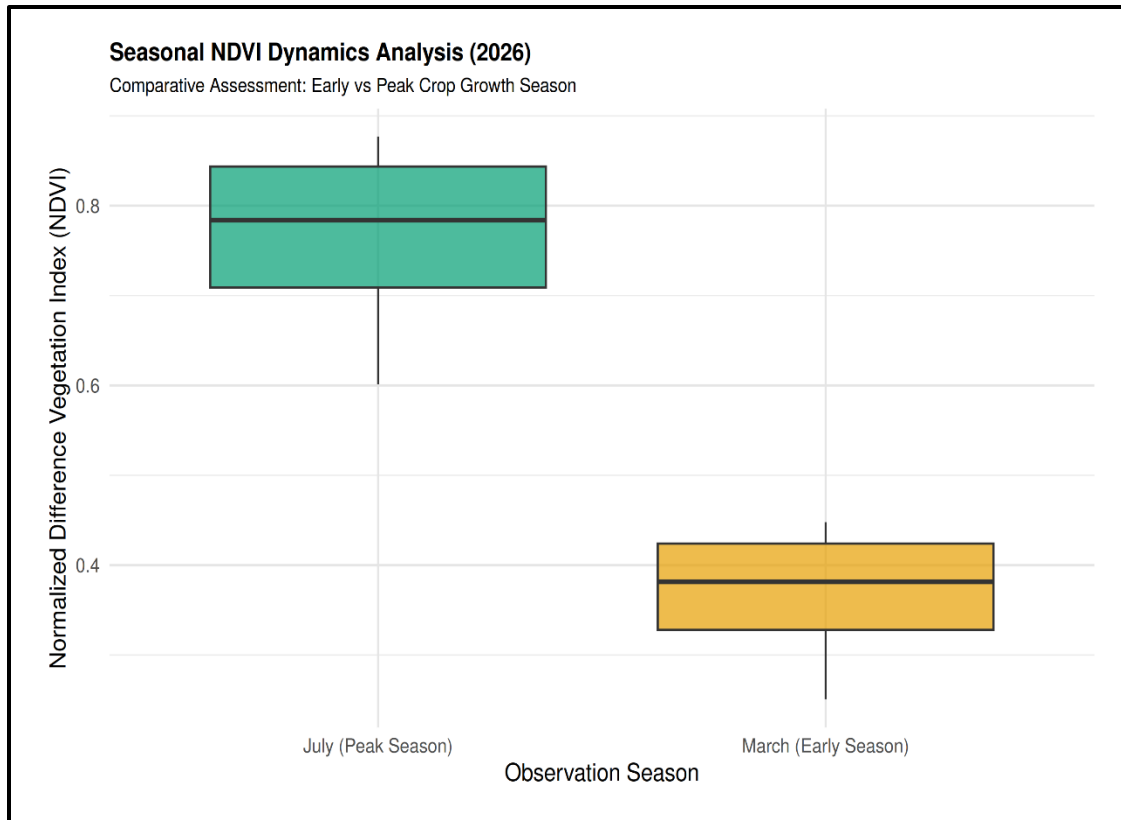


Figure 3: *Seasonal NDVI Boxplot comparing March (Early Season) vs July (Peak Season) canopy distribution.*

3. Spatial Pixel Correlation & Point Drift:

A scatter plot was constructed to evaluate how individual sampling locations tracked over time. The strong linear alignment confirms consistent spatial growth trends across all sampled plots.

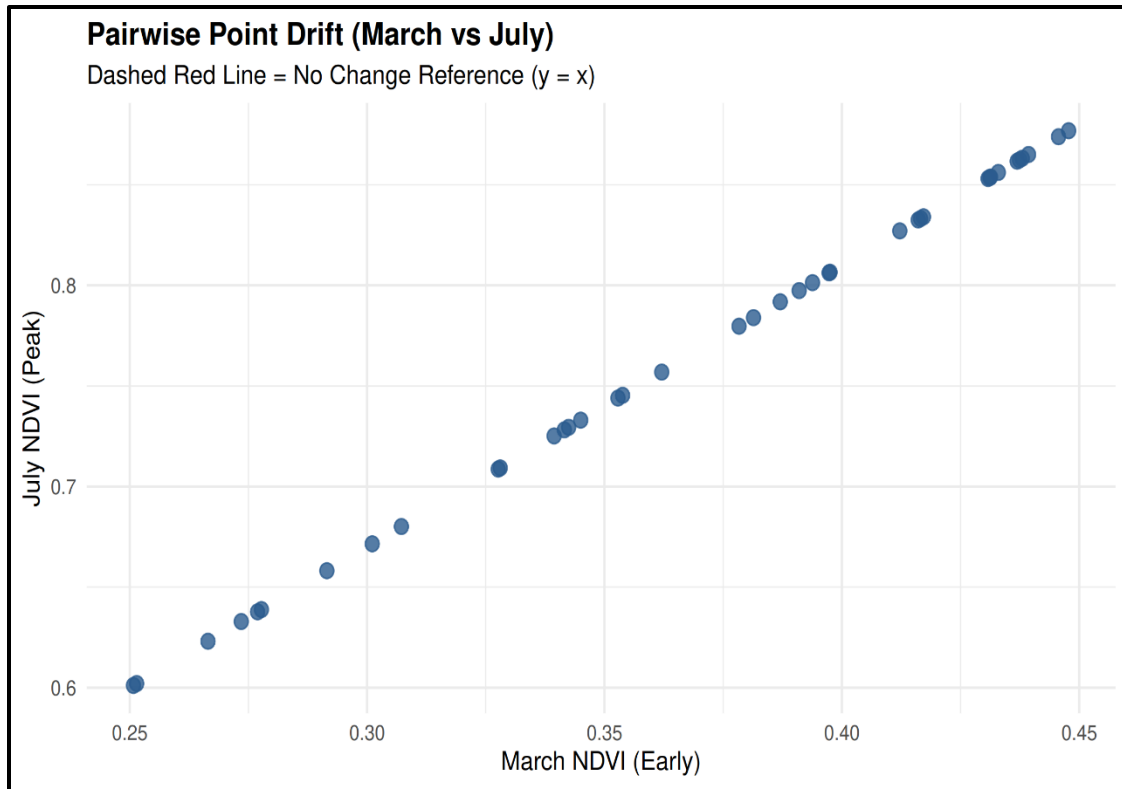


Figure 4: *Pairwise Point Drift Scatterplot evaluating per-pixel NDVI progression from March to July.*

4. Hypothesis Testing & Confidence Interval Estimation:

A paired t-test was conducted to confirm if the observed increase from early season ($Mean \approx 0.37$) to peak season ($Mean \approx 0.76$) was statistically significant:

- **March Mean NDVI:** ~ 0.37
- **July Mean NDVI:** ~ 0.76
- **p-value:** $p < 0.001$ (Statistically significant seasonal growth)

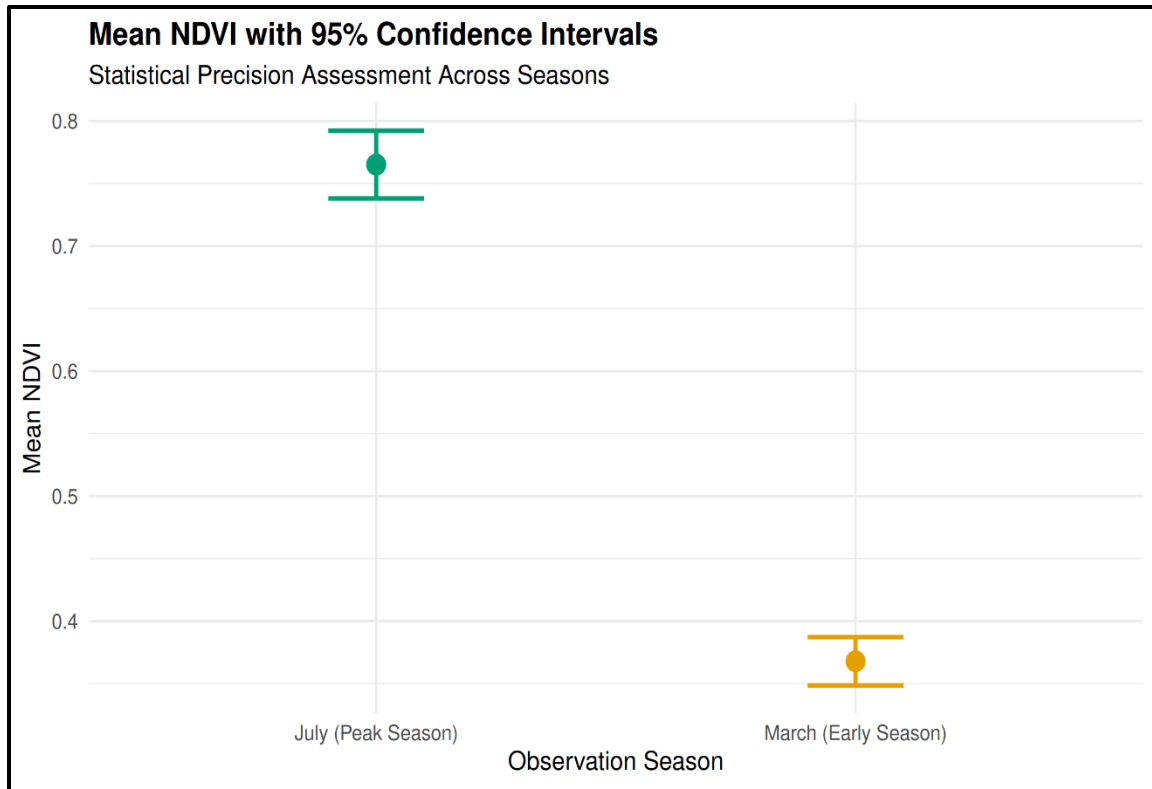


Figure 5: Mean NDVI with 95% Confidence Intervals confirming statistically significant seasonal growth performance.

5. Practical Business & Farming Value:

- **Variable-Rate Fertilization:** Farmers don't need to spray the whole field. They can focus fertilizer only on the yellow/red stress zones (saving 15–20% on input costs).
- **Targeted Irrigation:** Highlights exact dry patches caused by poor irrigation distribution.
- **Yield Prediction:** Gives an early warning 4 to 6 weeks before harvest so farmers know expected crop yields in advance.

6. Conclusion:

This project demonstrates how combining **Satellite Remote Sensing (QGIS)** with **Data Science (R)** transforms raw space imagery into practical, actionable farming insights. It proves that satellite analysis can accurately spot crop stress and track seasonal growth patterns with high statistical accuracy.